Appendix C: IEEE 830 Template

Software Requirements Specification

for

Gym Management System

Version 0.4

Prepared by Jaime Esteban Reaño - 905805

IFE – Software Engineering

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Revision History

Jaime Esteban Reaño	04/03/2026	Initial version of the SRS document created. Project topic selected: Gym Management System.	0.1
Jaime Esteban Reaño	11/03/2026	Added Introduction and Overall Description sections up to 2.1 based on requirements analysis.	0.2
Jaime Esteban Reaño	18/03/2026	Updated product features and user classes according to UML analysis activities.	0.3
Jaime Esteban Reaño	25/03/2026	Incorporated initial class diagram concepts and domain structure for the Gym Management System.	0.4

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1. Introduction

1.1. Purpose

This document specifies the software requirements for the Gym Management System, a software solution intended to support the daily management of a gym. The system will provide functionalities for member registration, class booking, trainer management, membership tracking, and payment control. This specification defines the current scope of the project and will be progressively updated as new analysis and design artifacts are developed during the course.

1.2. Document Conventions

Functional requirements are identified using the prefix FR, while non-functional requirements are identified using the prefix NFR. All mandatory requirements are written using the word “shall” to ensure precision and verifiability. UML diagrams included in the document are used as analysis and design support artifacts.

1.3. Intended Audience and Reading Suggestions

This document is intended for course instructors, project team members, testers, and future developers of the Gym Management System. Readers should begin with the Introduction and overall description sections to understand the purpose and scope of the project, and then continue with the system features and interface requirements

1.4. Product Scope

This Gym Management System aims to digitize and simplify the management of gym operations. Its main objective is to centralize information related to members, trainers, fitness classes, memberships, and payments in a single system. The expected benefits include improved administrative efficiency, reduced manual errors, easier scheduling, and better service for gym members

1.5. References

- Lab 1: Requirements Analysis – Software Engineering course materials

- Lab 2: UML Use Case Diagrams – Software Engineering course materials

- Lab 3: UML CLASS Diagrams – Software Engineering course materials

- IEEE 830 Software Requirements Specification Template, Appendix C course

-Lab 4: UML Sequence, State, and Activity Diagrams

2. Overall Description

2.1. Product Perspective

This Gym Management System is conceived as a new standalone software product. It is intended to replace informal or manual management methods based on spreadsheets, paper records, or disconnected applications. The system will centralize core gym operations in a single platform.

2.2. Product Features

- Member registration and profile management
- Membership plan management
- Trainer management
- Class scheduling and booking
- Payment tracking and membership validation
- Attendance monitoring

2.3. User Classes and Characteristics

The main user classes of the Gym Management System are Administrators, Trainers, and Members. Administrators are the primary management users and have the highest privilege level, being responsible for registering members, managing classes, assigning trainers, and controlling payments. Trainers use the system to consult assigned sessions, review schedules, and possibly track attendance. Members are end users who interact with the system to manage their personal information, check available classes, make reservations, and review membership status.

2.4. Operating Environment

This system is expected to operate as a web – based application. You could access it through modern browsers on desktop and also mobile devices. It should be compatible with common operating systems, like Windows, macOS, Android and even iOS , through browser access. However, an internet connection is required for its normal operation.

2.5. Design and Implementation Constraints

The project is developed in an academic environment and therefore its scope is limited to the core functionalities of a gym management platform. The system should prioritize clarity, usability, and modularity over advanced commercial features. Personal data related to members and trainers must be handled securely. The specification and design of the project will evolve incrementally according to the software engineering laboratory activities carried out during the course.

2.6. User Documentation

The system is expected to include basic user documentation adapted to an academic project context. This documentation may include a short administrator guide explaining how to register members, manage classes, assign trainers, and record payments, as well as a brief user guide for members describing how to view classes, make reservations, and review membership information. If implemented, online help or embedded interface hints may also be provided. The documentation will be delivered in digital format, most likely as PDF files and brief in-system textual guidance.

2.7. Assumptions and Dependencies

The specification assumes that the gym operates with clearly defined user roles, including administrators, trainers, and members. It is also assumed that the system will be accessed through devices with a stable internet connection and a compatible web browser. The project depends on decisions that may be taken in later stages of development, such as the database solution, the authentication mechanism, and the deployment environment. Since the project is developed incrementally in parallel with the laboratory activities of the course, some requirements and design decisions may evolve as new software engineering concepts are introduced.

3. System Features

3.1. Member Management

This feature covers the registration, storage, consultation, and update of member information. It also includes the management of membership status and related administrative data.

3.2 Trainer and Class Management

This feature supports the registration and management of trainers, as well as the creation and scheduling of fitness classes. It also includes the assignment of trainers to specific class sessions.

3.3 Booking and Attendance Management

This feature allows members to view available classes and book sessions when places are available. It also supports the recording or monitoring of attendance for scheduled activities.

3.3 Payment and Membership Control

This feature manages membership plans, payment records, and membership validity. It helps administrators verify whether a member has an active subscription and whether payments are up to date.

3.«N». «Additional feature blocks as needed»

4. External Interface Requirements

4.1. User Interfaces Overview

The Gym Management System will provide a graphical user interface (GUI) accessible via web browsers on desktop and mobile devices. Users will interact with the system through buttons, forms, and menus, entering data via text fields and selecting options from lists.

The system will offer different views depending on the user role (Administrator, Trainer, Member), allowing each user to perform their specific tasks such as managing data, viewing schedules, or booking classes.

The interface will provide clear feedback through confirmation and error messages, and will follow a consistent and user-friendly layout to ensure easy navigation and efficient use of the system.

4.2. Hardware Interfaces

The Gym Management System does not require any specific hardware interfaces, as it is designed to operate as a web-based application. Users will access the system through standard devices such as desktop computers, laptops, tablets, and smartphones.

The system interacts with hardware components indirectly through web browsers, using standard input and output devices such as keyboards, touchscreens, and displays. No specialized hardware or direct hardware communication protocols are required.

4.3. Software Interfaces

The Gym Management System will interact with standard software components required for a web-based application. This includes a web browser on the client side and a backend server responsible for processing requests and managing system logic.

The system will interface with a relational database management system (e.g., MySQL or PostgreSQL) to store and retrieve data related to members, trainers, classes, bookings, and payments. Data exchanged between the system and the database will include user information, schedules, and transaction records.

Communication between components will follow standard web protocols such as HTTP/HTTPS. The system may also rely on common software libraries and frameworks for handling authentication, data validation, and user interface rendering.

All data exchanged between components will be structured and consistent to ensure data integrity. No special data-sharing mechanisms are required beyond standard database operations and client-server communication.

5. System Features/Modules

5.1. Member Management

5.1.1 Description and Priority

This feature manages the registration, consultation, update, and storage of member information in the Gym Management System. It allows administrators to create and maintain member profiles and supports the control of basic personal and membership-related data.

Priority: High

5.1.2 Stimulus/Response Sequences

Stimulus: An administrator submits a new member registration form.

Response: The system validates the entered data and, if valid, creates a new member profile in the database.

Stimulus: An administrator requests to view a member profile.

Response: The system retrieves and displays the stored member information.

Stimulus: An administrator updates a member's personal or membership information.

Response: The system validates the new data and stores the updated information.

Stimulus: An administrator enters incomplete or invalid member data.

Response: The system rejects the operation and displays an error message indicating the incorrect or missing fields.

5.1.3 Functional Requirements

REQ-1.1: The system shall allow administrators to register a new member by entering personal and contact information.

REQ-1.2: The system shall validate mandatory fields before creating a member profile.

REQ-1.3: The system shall store each registered member with a unique identifier.

REQ-1.4: The system shall allow administrators to consult the profile information of any registered member.

REQ-1.5: The system shall allow administrators to update member data, including contact details and membership-related information.

REQ-1.6: The system shall reject the registration or update of a member if required data is missing or invalid.

REQ-1.7: The system shall display a confirmation message when a member profile is successfully created or updated.

REQ-1.8: The system shall preserve the consistency of stored member information after any modification.

5.2. Trainer and Class Management

5.2.1 Description and Priority

This feature supports the management of trainers and fitness classes. It allows administrators to register trainers, create class sessions, assign trainers to classes, and maintain schedules for gym activities.

Priority: High

5.2.2 Stimulus/Response Sequences

Stimulus: An administrator registers a new trainer in the system.

Response: The system validates the trainer data and stores the new trainer profile.

Stimulus: An administrator creates a new fitness class.

Response: The system stores the class information, including name, schedule, capacity, and assigned trainer if available.

Stimulus: An administrator assigns a trainer to a class session.

Response: The system links the trainer to the selected class and updates the schedule.

Stimulus: An administrator attempts to assign a trainer to overlapping class sessions.

Response: The system rejects the assignment and displays an error message.

5.2.3 Functional Requirements

REQ-2.1: The system shall allow administrators to register and store trainer information.

REQ-2.2: The system shall allow administrators to create fitness classes with attributes such as class name, date, time, duration, and capacity.

REQ-2.3: The system shall allow administrators to assign a trainer to a class session.

REQ-2.4: The system shall allow administrators to modify or cancel scheduled classes.

REQ-2.5: The system shall prevent assigning a trainer to two class sessions scheduled at overlapping times.

REQ-2.6: The system shall allow trainers to view the classes assigned to them.

REQ-2.7: The system shall display updated scheduling information after a class or trainer assignment is created or modified.

REQ-2.8: The system shall reject invalid class definitions, including missing schedule data or invalid capacity values.6. Nonfunctional Requirements

5.3. Booking and Attendance Management

5.3.1 Description and Priority

This feature allows members to consult available classes and reserve places in scheduled sessions. It also supports attendance monitoring for booked activities, helping trainers or administrators keep track of participation.

Priority: High

5.3.2 Stimulus/Response Sequences

Stimulus: A member requests to view available classes.

Response: The system displays the list of scheduled classes with available places.

Stimulus: A member books a place in a class.

Response: The system verifies availability and membership validity, then records the booking if the conditions are satisfied.

Stimulus: A member attempts to book a class with no available places.

Response: The system rejects the booking and informs the user that the class is full.

Stimulus: A trainer or administrator records attendance for a scheduled class.

Response: The system stores the attendance status for the corresponding members.

5.3.3 Functional Requirements

REQ-3.1: The system shall allow members to view the list of available classes and their schedules.

REQ-3.2: The system shall allow members to book a class only if places are available.

REQ-3.3: The system shall verify that a member has an active membership before confirming a booking.

REQ-3.4: The system shall reduce the number of available places when a booking is successfully completed.

REQ-3.5: The system shall reject a booking request if the class has reached its maximum capacity.

REQ-3.6: The system shall allow trainers or administrators to record attendance for scheduled classes.

REQ-3.7: The system shall store attendance information associated with both the member and the class session.

REQ-3.8: The system shall provide confirmation or error messages after each booking or attendance action.

5.4. Payment and Membership Control

5.4.1 Description and Priority

This feature manages membership plans, payment registration, and membership validity. It helps administrators verify whether members are up to date with payments and whether they are eligible to access services such as class booking.

Priority: High

5.4.2 Stimulus/Response Sequences

Stimulus: An administrator registers a payment for a member.

Response: The system stores the payment record and updates the member's membership status if applicable.

Stimulus: An administrator checks the membership validity of a member.

Response: The system displays the current membership plan, payment status, and validity period.

Stimulus: A member with an expired membership attempts to book a class.

Response: The system denies the booking and informs the member that the membership is inactive.

Stimulus: An administrator assigns or renews a membership plan.

Response: The system updates the member's plan and corresponding validity dates.

5.4.3 Functional Requirements

REQ-4.1: The system shall allow administrators to define and manage membership plans.

REQ-4.2: The system shall allow administrators to register payments made by members.

REQ-4.3: The system shall store the date, amount, and associated member for each payment record.

REQ-4.4: The system shall maintain the current membership status of each member.

REQ-4.5: The system shall allow administrators to renew or update a member's subscription plan.

REQ-4.6: The system shall identify whether a member has an active or expired membership.

REQ-4.7: The system shall prevent class booking when the member's membership is not active.

REQ-4.8: The system shall display payment and membership information when requested by authorized users.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The system shall display standard user interface pages within 3 seconds under normal operating conditions.

NF-1.2: The system shall process member registrations, bookings, and payment updates without noticeable delay for typical academic project usage.

NF-1.3: The system shall support concurrent access by multiple users with consistent system behavior.

6.2. Security

NF-2.1: The system shall require user authentication before granting access to protected functionalities.

NF-2.2: The system shall restrict access to functionalities according to the user role (Administrator, Trainer, or Member).

NF-2.3: The system shall protect personal and payment-related data from unauthorized access.

NF-2.4: The system shall use secure communication mechanisms such as HTTPS when deployed in a networked environment.

6.3. Usability

NF-3.1: The system shall provide a clear and consistent graphical user interface for all supported user roles.

NF-3.2: The system shall display understandable error and confirmation messages after user actions.

NF-3.3: The system shall be easy to learn for first-time users with basic digital skills.

6.4. Reliability

NF-4.1: The system shall preserve stored data correctly after valid operations.

NF-4.2: The system shall handle invalid input without crashing or corrupting stored information.

NF-4.3: The system shall maintain consistency between bookings, membership status, and class capacity data.

6.5. Maintainability

NF-5.1: The system shall be designed in a modular way to facilitate future modifications and extensions.

NF-5.2: The system shall maintain clear separation between interface, logic, and data management components.

NF-5.3: The code and documentation shall be understandable for future developers working on the project.

6.6. Portability

NF-6.1: The system shall be accessible through modern web browsers on desktop and mobile devices.

NF-6.2: The system shall be compatible with common operating systems such as Windows, macOS, Android, and iOS through browser access.